

STAT 240 Midterm

February 25th, 2026

Owen G. Ward

Please read all instructions on this page before beginning the exam.

1. This exam will start at 1:35pm. You will have 45 minutes to complete the exam, **which includes the time required to submit to crowdmark**. You can download the data (below) before that time.
2. **Any work not submitted by 2.20pm will receive 0. It is your responsibility to ensure that you correctly submit a pdf document by this time.**
3. You can use the slides from the class (pdf or html) and any notes you have created in R. If you have a web browser open, it can only be used to access crowdmark, create your pdf file, or to view the class slides.
4. You must submit a **rendered** pdf file for your exam to be graded. You can use the template file provided last week, which contains all packages needed. **Any submission not in this format will receive 0. Please give your Name clearly at the start of your pdf submission.**
5. There are three problems on this midterm, each containing multiple parts. Please make sure you do not miss any questions and clearly label each part of the problem in your submission.

Dataset

For this exam we will make use of the following dataset, `nhl_data.tsv`, which you can download from the internet using the following command:

```
download.file("https://owenward.github.io/files/nhl_data.tsv",  
             "nhl_data.tsv")
```

If you are unable to read or load this data you can use the following data (with an appropriate penalty):

```
nhl_data_fake <- data.frame(Team = c("LA", "STL", "EDM", "CAR"),  
                           W = c(47, 37, 50, 52),  
                           PTS = c(104, 81, 109, 113),  
                           GF = rpois(n = 4, lambda = 5),  
                           GA = rpois(n = 4, lambda = 4),  
                           DIFF = rpois(n = 4, lambda = 1),  
                           CONF = c("WESTERN", "EASTERN", "WESTERN", "EASTERN"))
```

Problem 1: Reading and Writing Data (6 points)

- (a) Read `nhl_data.tsv` into your R session with the appropriate commands based on the file format, and subset this data, keeping only the rows where the value of the column `W` is greater than 50. Store this subset data in an R object `nhl_sub`. Print out the number of rows in `nhl_sub`. **(2 points)**
- (b) Modify `nhl_sub` to remove the columns `GF`, `GA`, `DIFF`. Store this data, without these columns, in an R object called `nhl_mod`. **(2 points)**
- (c) Now, using `nhl_mod`, write this to a file `nhl.psv`, where the separator between the columns in this dataset is given by `._.` (dot, underscore, dot). **(2 points)**

Problem 2: SQL (7 points)

- (a) Using the original data in `nhl_data.tsv`, create an RSQL database, called `nhl.sqlite`, and write the data in the `tsv` file into this database, calling it `NHL`. **(2 points)**
- (b) Write a SQL command to select all rows from `NHL` where `W` is greater than 50. Confirm and show that this query returns the same number of rows as the corresponding R code you wrote in Problem 1a. **(3.5 points)**
- (c) Write a second SQL command to now group by the `CONF` variable and count the number of rows and the average number of wins (`W`), across these groups. Print the output of this SQL command. Make sure that your SQL command returns the `CONF` for each row in the output, and call this average `avg_wins`. You do not need to save this output, but you must show the output of the query in your submission. **(1.5 points)**

Problem 3: Linear Regression (7 points)

We want to try predict how many points (PTS) the teams in `nhl_data` have, based on the number of wins they have (W).

- (a) Construct a scatter plot of this data, showing the number of wins on the x-axis and the number of points on the y-axis. You don't need to worry about axis labels, etc. **(2 points)**
- (b) Fit a linear regression to this data and show the estimated intercept and slope from the fitted model. **(3 points)**
- (c) The estimates of the intercept and slope in the previous section minimize the sum of squared error for this data. Compute this sum of squared error and show that if you change either of the values of the intercept or slope, the sum of squares error will be larger. (That is, pick random values of both and compute the sum of squares error.) **(2 points)**